Available online at www.sciencedirect.com
ScienceDirect
Journal of Process Control 18 (2008) 769–779
JOURNAL OF PROCESS CONTROL
www.elsevier.com/locate/jprocont

A generalized autocovariance least-squares method for Kalman filter tuning

Bernt M. Åkesson [a,*], John Bagterp Jørgensen [b], Niels Kjølstad Poulsen [b], Sten Bay Jørgensen [a]

[a] *CAPEC, Department of Chemical Engineering, Technical University of Denmark, DK-2800 Lyngby, Denmark*
[b] *Informatics and Mathematical Modelling, Technical University of Denmark, DK-2800 Lyngby, Denmark*

Received 28 June 2007; received in revised form 31 October 2007; accepted 3 November 2007

Abstract

This paper discusses a method for estimating noise covariances from process data. In linear stochastic state-space representations the true noise covariances are generally unknown in practical applications. Using estimated covariances a Kalman filter can be tuned in order to increase the accuracy of the state estimates. There is a linear relationship between covariances and autocovariance. Therefore, the covariance estimation problem can be stated as a least-squares problem, which can be solved as a symmetric semidefinite least-squares problem. This problem is convex and can be solved efficiently by interior-point methods. A numerical algorithm for solving the symmetric is able to handle systems with mutually correlated process noise and measurement noise.
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Keywords: Kalman filtering; Covariance estimation; Optimal estimation; State estimation; Semidefinite programming

1. Introduction

In state estimation the system state vector is reconstructed from process measurements. State estimation has important applications in control, monitoring and fault detection of chemical processes. The Kalman filter and its counterpart for nonlinear systems, the extended Kalman filter, are well-established techniques for state estimation [12]. However, a well-known drawback of Kalman filters is that knowledge about process and measurement noise statistics is required from the user. In practical applications the noise covariances are generally not known. Tuning the filter, i.e. choosing the values of the process and measurement noise covariances such that the filter performance is optimized with respect to some performance index, is a challenging task. If performed manually in an ad hoc fashion it represents a considerable burden on the user. Therefore, there is a need for a tool that can perform filter tuning or and hence assist the user.

The filter tuning problem is essentially a covariance estimation problem and the Kalman filter gain is computed based on the estimated covariances. This issue has been addressed in numerous papers and a number of methods have been presented, cf. discussion in [18,17] and references therein. A promising technique for covariance estimation is the autocovariance least-squares method proposed recently by Odelson et al. for linear time-invariant systems [18]. This method is based on the estimated autocovariance of the output innovations, which is used to compute a least-squares estimate of the noise covariance matrices. The estimation problem can be stated in the form of a linear least-squares problem with additional constraints to ensure positive semidefiniteness of the covariance matrices. The problem can be stated in the form of a convex semidefinite program, which can be solved by interior-point methods.

* Corresponding author. Present address: Finnish Defence Forces Technical Research Centre, Electronics and Information Technology Division, Finland. Tel.: +358 19 181 3547.
E-mail addresses: bernt.akesson@mil.fi (B.M. Åkesson), jbj@imm.dtu.dk (J.B. Jørgensen), nkp@imm.dtu.dk (N.K. Poulsen), sbj@kt.dtu.dk (S.B. Jørgensen).

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doi:10.1016/j.jprocont.2007.11.003

This paper presents two contributions. The first contribution is the generalization of the autocovariance least-squares method to systems with correlated noise. Models used by model predictive controllers are typically identified from data. Such models may have correlated process and measurement noise. This holds in particular for models obtained by subspace identification [24], in which an innovations form is used. Moreover, as shown in [1], sampling a continuous-time system with uncorrelated noise may result in a system with correlated noise in discrete-time. The second contribution is an interior-point predictor–corrector algorithm for solving the symmetric semidefinite least-squares problem.

This paper is organized as follows. The problem is stated in Section 2. In Section 3, the method is outlined. Disturbance structure identification and rank minimization is discussed in Section 4. Algorithms for solving the semidefinite least-squares problem are discussed in Section 5. The tuning method is demonstrated on a numerical example in Section 6, followed by conclusions in Section 7.

2. Problem statement

We consider a linear state-space model in discrete-time

$$\begin{aligned}\bar{x}_{k+1} &= \Phi_x \bar{x}_k + \Phi_u u_k + \Phi_w w_k,\\ y_k &= \Gamma_x \bar{x}_k + v_k,\end{aligned} \tag{1}$$

where $\Phi_x \in \mathbb{R}^{n_x \times n_x}$, $\Phi_u \in \mathbb{R}^{n_x \times n_u}$, $\Phi_w \in \mathbb{R}^{n_x \times n_w}$ and $\Gamma_x \in \mathbb{R}^{n_y \times n_y}$. The process noise w_k and the measurement noise v_k are zero-mean white-noise processes with the following covariance structure:

$$\begin{bmatrix} w_k \\ v_k \end{bmatrix} \sim N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} Q_w & S_{wv} \\ S_{wv}^{\mathrm{T}} & R_v \end{bmatrix}\right). \tag{2}$$

To achieve offset-free control in model predictive control, an integrating disturbance model is added [20]. Here, an unstructured disturbance model is considered

$$\begin{aligned}\bar{x}_{k+1} &= \Phi_x \bar{x}_k + \Phi_u u_k + \Phi_d d_k + \Phi_w w_k,\\ d_{k+1} &= d_k + \xi_k,\\ y_k &= \Gamma_x \bar{x}_k + \Gamma_d d_k + v_k,\end{aligned} \tag{3}$$

where $\Phi_d \in \mathbb{R}^{n_x \times n_d}$ and $\Gamma_d \in \mathbb{R}^{n_y \times n_d}$. The white noise ξ_k in the disturbance model has covariance Q_ξ which is uncorrelated with the process noise and the measurement noise. The disturbance model can be a pure input disturbance model, in which case we have $\Phi_d = \Phi_u$ and $\Gamma_d = 0$, or a pure output disturbance by setting $\Phi_d = 0$ and $\Gamma_d = I$.

An augmented model is formed from (3)

$$\begin{aligned}x_{k+1} &= Ax_k + Bu_k + G\omega_k,\\ y_k &= Cx_k + v_k,\end{aligned} \tag{4}$$

where the following definitions have been introduced:

$$x_k = \begin{bmatrix} \bar{x}_k \\ d_k \end{bmatrix}, \quad \omega_k = \begin{bmatrix} w_k \\ \xi_k \end{bmatrix}$$

and

$$A = \begin{bmatrix} \Phi_x & \Phi_d \\ 0 & I \end{bmatrix}, \quad B = \begin{bmatrix} \Phi_u \\ 0 \end{bmatrix}, \quad G = \begin{bmatrix} \Phi_w & 0 \\ 0 & I \end{bmatrix},$$

$$C = [\Gamma_x \quad \Gamma_d].$$

We also define covariance matrices

$$\overline{Q}_w = \begin{bmatrix} Q_w & 0 \\ 0 & Q_\xi \end{bmatrix}, \quad \overline{S}_{wv} = [S_{wv} 0].$$

Assume that a steady-state Kalman filter is used to estimate the state. The filter can be either in the one-step predicting form:

$$\hat{x}_{k+1|k} = A\hat{x}_{k|k-1} + Bu_k + K_{\mathrm{p}}(y_k - C\hat{x}_{k|k-1}) \tag{5}$$

or in the filtering form, with time update

$$\hat{x}_{k+1|k} = A\hat{x}_{k|k} + Bu_k + G\hat{\omega}_{k|k} \tag{6}$$

and measurement update

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_{\mathrm{f}x}(y_k - C\hat{x}_{k|k-1}), \tag{7}$$

$$\hat{\omega}_{k|k} = K_{\mathrm{f}w}(y_k - C\hat{x}_{k|k-1}). \tag{8}$$

The Kalman filter gains K_{p} and $K_{\mathrm{f}x}$ are defined as

$$K_{\mathrm{p}} = (AP_{\mathrm{p}}C^{\mathrm{T}} + G\overline{S}_{wv})(CP_{\mathrm{p}}C^{\mathrm{T}} + R_v)^{-1}, \tag{9}$$

$$K_{\mathrm{f}x} = P_{\mathrm{p}}C^{\mathrm{T}}(CP_{\mathrm{p}}C^{\mathrm{T}} + R_v)^{-1}, \tag{10}$$

$$K_{\mathrm{f}w} = \overline{S}_{wv}(CP_{\mathrm{p}}C^{\mathrm{T}} + R_v)^{-1} \tag{11}$$

and P_{p} is the covariance of the state prediction error, $\tilde{x}_{k|k-1} = x_k - \hat{x}_{k|k-1}$. The covariance $P_{\mathrm{p}} = E[\tilde{x}_{k|k-1}\tilde{x}_{k|k-1}^{\mathrm{T}}]$ is obtained as the solution to the Riccati equation

$$\begin{aligned}P_{\mathrm{p}} &= AP_{\mathrm{p}}A^{\mathrm{T}} + G\overline{Q}_w G^{\mathrm{T}}\\ &\quad - (AP_{\mathrm{p}}C^{\mathrm{T}} + G\overline{S}_{wv})(CP_{\mathrm{p}}C^{\mathrm{T}} + R_v)^{-1}(CP_{\mathrm{p}}A^{\mathrm{T}} + \overline{S}_{wv}^{\mathrm{T}}G^{\mathrm{T}}).\end{aligned} \tag{12}$$

In the following approach it is assumed that process data and the model are given, along with an initial suboptimal Kalman filter, based on initial guesses $Q_{w,0}$, $R_{v,0}$, $S_{wv,0}$ and $Q_{\xi,0}$. The objective is to estimate the covariance matrices Q_w, R_v, S_{wv} and Q_ξ and use these to compute the Kalman filter gains K_{p} or $K_{\mathrm{f}x}$ and $K_{\mathrm{f}w}$.

3. Autocovariance least-squares estimation

An extension of the autocovariance least-squares method for covariance estimation presented in [18,17] is outlined in this section. A general state-space model for the prediction error can be defined by combining either (4) and (5) or (4), (6) and (7). Regardless of which form of the filter is used, the model can be written in the general form

$$\begin{aligned}\tilde{x}_{k+1|k} &= \overline{A}\tilde{x}_{k|k-1} + \overline{G}\bar{w}_k,\\ e_k &= \overline{C}\tilde{x}_{k|k-1} + \overline{H}v_k\end{aligned} \tag{13}$$

in which

$$\bar{w}_k = \begin{bmatrix} w_k \\ \xi_k \\ v_k \end{bmatrix}$$

and

$$\overline{A} = A - K_\mathrm{p} C, \quad \overline{G} = [\, G \quad -K_\mathrm{p} \,].$$

Note that e_k, $\overline{C}$ and $\overline{H}$ have different definitions depending on which form of the Kalman filter is used, as defined in Table 1.

The autocovariance of the prediction error is given by

$$R_{\mathrm{e},0} = E[e_k e_k^\mathrm{T}] = \overline{C} P_\mathrm{p} \overline{C}^\mathrm{T} + \overline{H} R_v \overline{H}^\mathrm{T}, \tag{14}$$

$$R_{\mathrm{e},j} = E[e_{k+j} e_k^\mathrm{T}] = \overline{CA}^j P_\mathrm{p} \overline{C}^\mathrm{T} + \overline{CA}^{j-1} V_x S_{wv} \overline{H}^\mathrm{T} - \overline{CA}^{j-1} K_\mathrm{p} R_v \overline{H}^\mathrm{T}, \quad j \geqslant 1 \tag{15}$$

in which we have introduced

$$V_x = \begin{bmatrix} \Phi_w \\ 0 \end{bmatrix}.$$

The theoretical autocovariance matrix is defined as

$$\mathscr{R}_\mathrm{e}(L) = \begin{bmatrix} R_{\mathrm{e},0} & R_{\mathrm{e},1}^\mathrm{T} & \cdots & R_{\mathrm{e},L-1}^\mathrm{T} \\ R_{\mathrm{e},1} & R_{\mathrm{e},0} & & R_{\mathrm{e},L-2}^\mathrm{T} \\ \vdots & & \ddots & \vdots \\ R_{\mathrm{e},L-1} & R_{\mathrm{e},L-2} & \cdots & R_{\mathrm{e},0} \end{bmatrix}. \tag{16}$$

This matrix is block-diagonal for an ideal filter. Substitution of (14) and (15) into (16) followed by separation of terms is performed. In order to write the resulting equation more compactly we use the direct sum, defined as a block-diagonal matrix such that

$$\bigoplus_{i=1}^{L} A_i = \begin{bmatrix} A_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & A_L \end{bmatrix}.$$

The theoretical autocovariance matrix can then be written as

$$\begin{aligned} \mathscr{R}_\mathrm{e}(L) = {} & \overline{O} P_\mathrm{p} \overline{O}^\mathrm{T} + Z \left[\bigoplus_{i=1}^{L} Q_w \right] Z^\mathrm{T} + Z_d \left[\bigoplus_{i=1}^{L} Q_\xi \right] Z_d^\mathrm{T} \\ & + Z \left[\bigoplus_{i=1}^{L} S_{wv} \right] \Psi^\mathrm{T} + \Psi \left[\bigoplus_{i=1}^{L} S_{wv}^\mathrm{T} \right] Z^\mathrm{T} \\ & + \Psi \left[\bigoplus_{i=1}^{L} R_v \right] \Psi^\mathrm{T}, \end{aligned} \tag{17}$$

Table 1
Definitions of symbols in (13) for each Kalman filter form

Symbol	Predicting form	Filtering form
e_k	$y_k - C\hat{x}_{k\|k-1}$	$y_k - C\hat{x}_{k\|k}$
$\overline{C}$	C	$C - CK_\mathrm{fx} C$
$\overline{H}$	I	$I - CK_\mathrm{fx}$

where

$$Z = \Gamma \left[\bigoplus_{i=1}^{L} V_x \right], \quad Z_d = \Gamma \left[\bigoplus_{i=1}^{L} V_d \right], \quad \Psi = \Gamma \left[\bigoplus_{i=1}^{L} (-K_\mathrm{p}) \right] + \bigoplus_{i=1}^{L} \overline{H}$$

and

$$\overline{O} = \begin{bmatrix} \overline{C} \\ \overline{CA} \\ \vdots \\ \overline{CA}^{L-1} \end{bmatrix}, \quad \Gamma = \begin{bmatrix} 0 & \cdots & 0 & 0 & 0 \\ \overline{C} & & 0 & 0 & 0 \\ \overline{CA} & & 0 & 0 & 0 \\ \vdots & \ddots & & & \vdots \\ \overline{CA}^{L-2} & \cdots & \overline{CA} & \overline{C} & 0 \end{bmatrix},$$

$$V_d = \begin{bmatrix} 0 \\ I_{n_d} \end{bmatrix}.$$

We apply the vec operator to (17) in order to state the problem as a linear least-squares problem. The vec operator stacks the matrix columns to form a column matrix [3]. The covariance of the state prediction error P_p can be eliminated from the final expression, by deriving an analytical solution to a Lyapunov equation. The derivation is presented in Appendix A.

$$\begin{aligned} \mathrm{vec}(P_\mathrm{p}) = {} & (I_{(n_x+n_d)^2} - \overline{A} \otimes \overline{A})^{-1} \big((V_x \otimes V_x) \mathrm{vec}(Q_w) \\ & + (V_d \otimes V_d) \mathrm{vec}(Q_\xi) - (V_x \otimes K_\mathrm{p}) \mathrm{vec}(S_{wv}^\mathrm{T}) \\ & - (K_\mathrm{p} \otimes V_x) \mathrm{vec}(S_{wv}) + (K_\mathrm{p} \otimes K_\mathrm{p}) \mathrm{vec}(R_v) \big). \end{aligned} \tag{18}$$

We now introduce three permutation matrices. The first one is used to replace the stacked direct sum with a simpler form that allows us to extract the covariance matrices. For an $m \times n$ matrix A we define a permutation matrix $U_{m,n,L}$, which is an $mnL^2 \times mn$ matrix of zeros and ones satisfying

$$\mathrm{vec}\left(\bigoplus_{i=1}^{L} A \right) = U_{m,n,L} \, \mathrm{vec}(A). \tag{19}$$

For a square matrix of size $p \times p$ we have the permutation matrix $U_{p,L} = U_{p,p,L}$.

Applying the vec operator to (17) yields

$$\begin{aligned} \mathrm{vec}(\mathscr{R}_\mathrm{e}(L)) = {} & [(Z \otimes Z) U_{n_w,L} + D(V_x \otimes V_x)] \mathrm{vec}(Q_w) \\ & + [(Z_d \otimes Z_d) U_{n_d,L} \\ & + D(V_d \otimes V_d)] \mathrm{vec}(Q_\xi) + [(\Psi \otimes \Psi) U_{n_y,L} \\ & + D(K_\mathrm{p} \otimes K_\mathrm{p})] \mathrm{vec}(R_v) \\ & + [(\Psi \otimes Z) U_{n_w,n_y,L} - D(K_\mathrm{p} \otimes V_x)] \mathrm{vec}(S_{wv}) \\ & + [(Z \otimes \Psi) U_{n_y,n_w,L} - D(V_x \otimes K_\mathrm{p})] \mathrm{vec}(S_{wv}^\mathrm{T}) \end{aligned} \tag{20}$$

in which

$$D = (\overline{O} \otimes \overline{O}) \left(I_{(n_x+n_d)^2} - \overline{A} \otimes \overline{A} \right)^{-1}. \tag{21}$$

Eq. (20) can be stated compactly as

$$\mathrm{vec}(\mathscr{R}_\mathrm{e}(L)) = A_{LS} \, \mathrm{vec}(X) \tag{22}$$

with X defined as

$$X = \begin{bmatrix} Q_w & 0 & S_{wv} \\ 0 & Q_\xi & 0 \\ S_{wv}^{\mathrm{T}} & 0 & R_v \end{bmatrix}.$$

The parameter matrix A_{LS} is given by

$$A_{LS} = [A_Q \quad A_{Q_\xi} \quad A_R \quad A_S \quad A_{S^{\mathrm{T}}}]T, \tag{23}$$

where

$$\begin{aligned} A_Q &= (Z \otimes Z)U_{n_w,L} + D(V_x \otimes V_x), \\ A_{Q_\xi} &= (Z_d \otimes Z_d)U_{n_d,L} + D(V_d \otimes V_d), \\ A_R &= (\Psi \otimes \Psi)U_{n_y,L} + D(K_{\mathrm{p}} \otimes K_{\mathrm{p}}), \\ A_S &= (\Psi \otimes Z)U_{n_w,n_y,L} - D(K_{\mathrm{p}} \otimes V_x), \\ A_{S^{\mathrm{T}}} &= (Z \otimes \Psi)U_{n_y,n_w,L} - D(V_x \otimes K_{\mathrm{p}}) \end{aligned} \tag{24}$$

and T is a permutation matrix given by

$$T = \begin{bmatrix} [I_{n_w} \quad 0 \quad 0] \otimes [I_{n_w} \quad 0 \quad 0] \\ [0 \quad I_{n_d} \quad 0] \otimes [0 \quad I_{n_d} \quad 0] \\ [0 \quad 0 \quad I_{n_y}] \otimes [0 \quad 0 \quad I_{n_y}] \\ [0 \quad 0 \quad I_{n_y}] \otimes [I_{n_w} \quad 0 \quad 0] \\ [I_{n_w} \quad 0 \quad 0] \otimes [0 \quad 0 \quad I_{n_y}] \end{bmatrix}. \tag{25}$$

Eq. (22) is a theoretical model of how process and measurement noise covariances will affect the autocovariance. However, the autocovariance of the prediction error can be estimated, leaving the noise covariance matrices as an unknown quantity, which can be estimated by solving a linear least-squares problem:

$$\min_X \frac{1}{2}\|A_{LS}\,\mathrm{vec}(X) - \mathrm{vec}(\widehat{\mathscr{R}}_{\mathrm{e}}(L))\|_2^2. \tag{26}$$

The autocovariance matrix $\widehat{\mathscr{R}}_{\mathrm{e}}(L)$ can be computed from process data. Given a sequence of data $\{e_i\}_{i=1}^{N_{\mathrm{d}}}$, the estimated autocovariance can be computed by

$$\widehat{R}_{\mathrm{e},j} = \frac{1}{N_{\mathrm{d}} - j}\sum_{i=1}^{N_{\mathrm{d}}-j} e_{i+j}e_i^{\mathrm{T}}, \tag{27}$$

where N_{d} is the length of the data sequence. The estimated autocovariance matrix $\widehat{\mathscr{R}}_{\mathrm{e}}(L)$ can then be formed analogously to (16).

However, solving (26) for X as a linear least-squares problem does not guarantee that the estimated covariance matrices are positive semidefinite. Furthermore, the parameter matrix A_{LS} may be poorly conditioned, which affects the accuracy of the solution. This can be remedied by adding to the objective function in (26) a regularization term, which penalizes deviations from the a priori guess. The estimation problem can be formulated as follows:

$$\begin{aligned} \min_X \quad & \left\{ \frac{1}{2}\underbrace{\|A_{LS}\,\mathrm{vec}(X) - \mathrm{vec}(\widehat{\mathscr{R}}_{\mathrm{e}}(L))\|_2^2}_{\Phi} + \frac{1}{2}\lambda\underbrace{\|\mathrm{vec}(X - X_0)\|_2^2}_{\Phi_r} \right\} \\ \text{s.t.} \quad & X \text{ symmetric positive semidefinite,} \end{aligned} \tag{28}$$

where λ is a regularization parameter chosen by the user which introduces a suitable bias-variance trade-off in the covariance estimation and

$$X_0 = \begin{bmatrix} Q_{w,0} & 0 & S_{wv,0} \\ 0 & Q_{\xi,0} & 0 \\ S_{wv,0}^{\mathrm{T}} & 0 & R_{v,0} \end{bmatrix}$$

are the covariances used to compute the initial Kalman filter gain. Eq. (28) constitutes a semidefinite least-squares problem, which is convex and can be solved by an interior-point method [14,18]. Algorithms for solving (28) are presented in Section 5. A suitable value for λ can be found e.g. using the L-curve method, in which (28) is solved for different values of λ and Φ_r is plotted versus Φ [13]. The λ value corresponding to the bend in the curve is then chosen. The optimal Kalman filter gain can then be computed from the estimated covariances using (9) or (10) after solving the Riccati equation (12).

Note that the process noise covariance Q_w and covariance Q_ξ have the same role in that they capture model mismatch and unknown disturbances. When estimating both, it is necessary to set parameter ρ to a large value so that Q_ξ will be used instead of Q_w for disturbance identification. This will be shown in Section 6.

4. Identification of disturbance structure

This section follows the procedure proposed by Rajamani and Rawlings [21,22]. There are usually only a few independent disturbances affecting the process state. Moreover, the noise shaping matrix Φ_w as seen in (1) may be unknown. In this case it is still possible to estimate $\Phi_w Q_w \Phi_w^{\mathrm{T}}$ but the solution will not be unique. In this case, minimization of the rank will be an additional objective in the least-squares problem. Solving the rank minimization problem is difficult and is therefore replaced by the matrix trace [9,8]. This follows from the result that a symmetric positive semidefinite matrix variable, which is the solution to the rank minimization problem, and is bounded by M, has a rank p_{rmp} that satisfies

$$p_{\mathrm{rmp}} \geqslant \frac{1}{M}p_{\mathrm{tr}},$$

where p_{tr} is the matrix trace [9].

In order to use incorporate the rank minimization term into the estimation problem we express the matrix trace as

$$\mathrm{Tr}(Q_w) = P\,\mathrm{vec}(X), \tag{29}$$

where P is a permutation matrix, which extracts and sums the diagonal elements of Q_w. The least-squares problem is now stated as

$$\min_X \quad \left\{\frac{1}{2}\left\|A_{LS}\,\mathrm{vec}(X) - \mathrm{vec}(\widehat{\mathscr{R}}_e(L))\right\|_2^2 + \frac{1}{2}\lambda\|\mathrm{vec}(X - X_0)\|_2^2 + \rho P\,\mathrm{vec}(X)\right\}$$
$$\text{s.t.} \quad X \text{ symmetric positive semidefinite.} \tag{30}$$

We can regard both λ and ρ as regularization parameters. For a fixed value of λ, the value of ρ can be chosen by using the L-curve method as shown in [21,22]. However, selection of multiple regularization parameters is a topic of ongoing research and has been studied e.g. in [2].

After solving (30) for an optimal value of ρ the minimal noise representation can be obtained from the estimated covariances through singular value decomposition (SVD). This will also give an estimate of the noise shaping matrix Φ_w. Covariance matrix Q_ξ has been omitted in the following, since ξ_k is assumed to be uncorrelated with process noise w_k and measurement noise v_k. Singular value decomposition of the estimated covariance matrix gives

$$\begin{bmatrix} \Phi_w Q_w \Phi_w^{\mathrm{T}} & \Phi_w S_{wv} \\ S_{wv}^{\mathrm{T}} \Phi_w^{\mathrm{T}} & R_v \end{bmatrix} = [\,G_1 \quad G_2\,] \begin{bmatrix} \Sigma_1 & 0 \\ 0 & \Sigma_2 \end{bmatrix} \begin{bmatrix} G_1^{\mathrm{T}} \\ G_2^{\mathrm{T}} \end{bmatrix}. \tag{31}$$

The singular values are separated into two parts: nonzero singular values and approximately zero singular values. This gives the number of disturbances affecting the system. The tolerance has to be specified by the user. If $\Sigma_2 \approx 0$, then

$$\begin{bmatrix} \Phi_w Q_w \Phi_w^{\mathrm{T}} & \Phi_w S_{wv} \\ S_{wv}^{\mathrm{T}} \Phi_w^{\mathrm{T}} & R_v \end{bmatrix} \approx G_1 \Sigma_1 G_1^{\mathrm{T}}.$$

By splitting G_1 into two parts

$$G_1 = \begin{bmatrix} G_{11} \\ G_{12} \end{bmatrix},$$

we can then write the system similarly to (1)

$$\begin{aligned} \bar{x}_{k+1} &= \Phi_x \bar{x}_k + \Phi_u u_k + G_{11}\underline{\omega}_k, \\ y_k &= \Gamma_x \bar{x}_k + G_{12}\underline{\omega}_k, \end{aligned} \tag{32}$$

where $\underline{\omega}_k$ is white noise with covariance Σ_1.

5. Semidefinite least-squares problem

5.1. Problem statement

We have the following optimization problem:

$$\min_X \quad \left\{\frac{1}{2}\|A\,\mathrm{vec}(X) - b\|_2^2 + \frac{1}{2}\lambda\|\mathrm{vec}(X - X_0)\|_2^2 + \rho P\,\mathrm{vec}(X)\right\}$$
$$\text{s.t.} \quad X = X^{\mathrm{T}} \succeq 0, \tag{33}$$

where X is a $n \times n$ matrix. This is a type of semidefinite programming has been discussed in [25]. Here we present an interior-point algorithm for solving the symmetric semidefinite least-squares problem (33). Algorithms have been developed for the related problem of solving

$$\min_X \quad \frac{1}{2}\|AX - B\|_{\mathrm{F}}^2$$
$$\text{s.t.} \quad X = X^{\mathrm{T}} \succeq 0, \tag{34}$$

where $\|\cdot\|_{\mathrm{F}}$ denotes the Frobenius norm [6,14]. It can be shown that (34) can be taken as a special case of (33). These algorithms can be extended to the problem of solving (28) with minor modifications.

We introduce a logarithmic barrier function $F(X)$, which enforces positive definiteness of X

$$F(X) = \begin{cases} -\ln\det X, & \text{if } X \succ 0, \\ +\infty, & \text{otherwise.} \end{cases} \tag{35}$$

Using the approach of Fiacco and McCormick [10], the least-squares problem (33) can then be expressed as the following unconstrained problem:

$$\phi = \min_X \left\{\frac{1}{2}\|A\,\mathrm{vec}(X) - b\|_2^2 + \frac{1}{2}\lambda\|\mathrm{vec}(X - X_0)\|_2^2 + \rho P\,\mathrm{vec}(X) + \tau F(X)\right\}. \tag{36}$$

Differentiation of (36) gives us

$$\nabla_{\mathrm{vec}(X)}\phi = (A^{\mathrm{T}}A + \lambda I)\mathrm{vec}(X) - A^{\mathrm{T}}b - \lambda\,\mathrm{vec}(X_0) + \rho P^{\mathrm{T}} - \tau\,\mathrm{vec}(X^{-1}). \tag{37}$$

We introduce matrix $2Y = \tau X^{-1}$. Y has to be symmetric and we need to include this condition. We then have

$$\frac{1}{2}(Y + Y^{\mathrm{T}}) = Y \iff \frac{1}{2}(I + K_{nn})\mathrm{vec}(Y) = \mathrm{vec}(Y). \tag{38}$$

Here, K_{nn} is the vec permutation matrix [15] such that for an $n \times n$ matrix A

$$\mathrm{vec}(A) = K_{nn}\,\mathrm{vec}(A^{\mathrm{T}}).$$

By combining (37) and (38) and defining matrix M_A and vector M_B

$$M_A = (I + K_{nn})(A^{\mathrm{T}}A + \lambda I),$$
$$M_B = (I + K_{nn})(A^{\mathrm{T}}b + \lambda\,\mathrm{vec}(X_0) - \rho P^{\mathrm{T}}),$$

we have the following necessary and sufficient conditions for an optimal solution of (36):

$$M_A\,\mathrm{vec}(X) - M_B - 2\,\mathrm{vec}(Y) = 0, \tag{39}$$
$$2YX - \tau I = 0, \tag{40}$$
$$X, Y \succ 0. \tag{41}$$

It is necessary to replace $2YX = \tau I$ with an equivalent symmetric equation $XY + YX = \tau I$, to force YX to be symmetric. Conditions (39)–(41) can then be expressed as

$$M_A\,\mathrm{vec}(X) - M_B - 2\,\mathrm{vec}(Y) = 0, \tag{42}$$
$$(I \otimes Y)\mathrm{vec}(X) + (I \otimes X)\mathrm{vec}(Y) - \tau\,\mathrm{vec}(I) = 0, \tag{43}$$
$$X, Y \succ 0. \tag{44}$$

If the point (X, Y) satisfies (42)–(44), then it is a primal–dual feasible point with duality gap

$$\mu = \langle X, Y\rangle = \mathrm{Tr}(XY) = n\tau, \tag{45}$$

which is a measure of the optimality of the current iterate (X, Y).

5.2. Predictor–corrector algorithm

For solving the symmetric positive semidefinite least-squares problem we have implemented a predictor–corrector algorithm. This algorithm can be viewed as an extension of an interior-point method for solving quadratic programming (QP) problems. An advantage of the predictor–corrector algorithm over a simple Newton-based one is that small step lengths are avoided [16]. A description of such an interior-point method can be found in [11].

The predictor–corrector algorithm alternates between affine (predictor) steps and predictor–corrector steps. The affine step is a pure Newton step

$$\begin{bmatrix} M_A - I_{n^2} & \\ Y \oplus Y & X \oplus X \end{bmatrix} \begin{bmatrix} \mathrm{vec}(\Delta X^{\mathrm{aff}}) \\ \mathrm{vec}(\Delta Y^{\mathrm{aff}}) \end{bmatrix} = -\begin{bmatrix} M_A - I_{n^2} & \\ I \otimes Y & I \otimes X \end{bmatrix} \begin{bmatrix} \mathrm{vec}(X) \\ \mathrm{vec}(Y) \end{bmatrix} + \begin{bmatrix} M_B \\ 0 \end{bmatrix}, \tag{46}$$

where the Kronecker sum [3] $A \oplus B = A \otimes I + I \otimes B$ is used for brevity. Symmetry of the step is maintained by setting

$$\Delta X^{\mathrm{aff}} \leftarrow (\Delta X^{\mathrm{aff}} + (\Delta X^{\mathrm{aff}})^{\mathrm{T}})/2,$$
$$\Delta Y^{\mathrm{aff}} \leftarrow (\Delta Y^{\mathrm{aff}} + (\Delta Y^{\mathrm{aff}})^{\mathrm{T}})/2.$$

The step length θ^{aff} is determined such that the affine iterate

$$X^{\mathrm{aff}} = X + \theta^{\mathrm{aff}} \Delta X^{\mathrm{aff}}, \tag{47}$$
$$Y^{\mathrm{aff}} = Y + \theta^{\mathrm{aff}} \Delta Y^{\mathrm{aff}} \tag{48}$$

satisfies $X^{\mathrm{aff}}, Y^{\mathrm{aff}} \succ 0$. The step length θ^{aff} is given by

$$\theta^{\mathrm{aff}} = \min\{1, c\theta_{\max}\},$$

where the maximum possible step size $\theta_{\max}$ is the minimum of the individual step sizes θ_X and θ_Y

$$\theta_{\max} = \min\{\theta_X, \theta_Y\}.$$

The step length parameter c is chosen adaptively, following a recommendation in [23]:

$$c = 0.9 + 0.09\theta_{r-1},$$

where θ_{r-1} is the step length used in the previous iteration. The initial value for the step length is $\theta_0 = 0$.

The individual step sizes θ_X and θ_Y are both chosen in a similar manner. Step length θ_X is the smallest positive solution such that

$$\det(X + \theta_X \Delta X) = 0.$$

If no positive solutions exist, then $\theta_X = \infty$. Here, θ_X is obtained as the minimum positive eigenvalue of a generalized eigenvalue problem:

$$Xv = \lambda(-\Delta X)v.$$

Alternatively, one can find the maximum eigenvalues of the problem

$$-\Delta X v = \lambda X v$$

and let

$$\theta_X = \begin{cases} 1/\lambda, & \text{if } \lambda > 0, \\ \infty, & \text{otherwise.} \end{cases}$$

In the same manner, θ_Y is determined. This guarantees that the affine iterate $(X^{\mathrm{aff}}, Y^{\mathrm{aff}})$ satisfies $X^{\mathrm{aff}}, Y^{\mathrm{aff}} \succ 0$.

The barrier parameter τ is given by

$$\tau = \sigma\hat{\mu}. \tag{49}$$

We define the normalized duality gap $\hat{\mu}$ as

$$\hat{\mu} = \langle X, Y \rangle / n = \mathrm{Tr}(XY)/n. \tag{50}$$

The centering parameter σ is updated using the Mehrotra rule

$$\sigma = \left(\frac{\hat{\mu}^{\mathrm{aff}}}{\hat{\mu}}\right)^3, \tag{51}$$

where $\hat{\mu}$ is computed by (50) and the normalized duality gap $\hat{\mu}^{\mathrm{aff}}$ for the affine iterate is

$$\hat{\mu}^{\mathrm{aff}} = \langle X^{\mathrm{aff}}, Y^{\mathrm{aff}} \rangle / n. \tag{52}$$

The predictor–corrector step is then given by

$$\begin{bmatrix} M_A & -I_{n^2} \\ Y \oplus Y & X \oplus X \end{bmatrix} \begin{bmatrix} \mathrm{vec}(\Delta X) \\ \mathrm{vec}(\Delta Y) \end{bmatrix} = -\begin{bmatrix} M_A & -I_{n^2} \\ I \otimes Y & I \otimes X \end{bmatrix} \begin{bmatrix} \mathrm{vec}(X) \\ \mathrm{vec}(Y) \end{bmatrix} + \begin{bmatrix} M_B \\ \mathrm{vec}(\tau I - \Delta X^{\mathrm{aff}} \Delta Y^{\mathrm{aff}} - \Delta Y^{\mathrm{aff}} \Delta X^{\mathrm{aff}}) \end{bmatrix}. \tag{53}$$

Again symmetry of the predictor–corrector step is maintained and the new iterate is chosen as

$$(\overline{X}, \overline{Y}) = (X, Y) + \theta(\Delta X, \Delta Y). \tag{54}$$

The step length θ is chosen using the same procedure as when choosing θ^{aff}.

Given a tolerance ϵ, an iterate $(\overline{X}, \overline{Y})$ is the ϵ-approximate solution to the semidefinite least-squares problem if it satisfies

The algorithm terminates successfully if the following criteria are met:

$$\|M_A \mathrm{vec}(\overline{X}) - M_B - 2\,\mathrm{vec}(\overline{Y})\|_\infty \leqslant tol_L \|(M_A, M_B)\|_\infty, \tag{55}$$
$$\hat{\mu} \leqslant tol_\mu. \tag{56}$$

These are analogous to the stopping criteria in [11]. The predictor–corrector algorithm is summarized in Fig. 1.

6. Numerical example: Van der Vusse reaction system

A non-isothermal Van der Vusse reaction scheme is used to demonstrate the Kalman filter tuning procedure. The Van der Vusse reaction has been used as a benchmark in several controller benchmark studies [5,19,4].

Algorithm 1 (Predictor-corrector algorithm)
Given: initial guesses X_0, Y_0 $(\in \mathbb{R}^{n\times n}) \succ 0$, tolerances tol_L, tol_μ and maximum number of iterations $MaxIt \geq 0$

Assign $(X,Y) \leftarrow (X_0,Y_0)$, $r \leftarrow 0$, $\theta_0 \leftarrow 0$, $tol_1 \leftarrow tol_L \|(M_A,M_B)\|_\infty$, $tol_2 \leftarrow tol_\mu$

while $(\|M_A \mathrm{vec}(X) - M_B - 2\mathrm{vec}(Y)\|_\infty > tol_1$ **or** $\langle X,Y\rangle/n > tol_2)$ **and** $r < MaxIt$ **do**
(1) Compute the affine direction $(\Delta X^{\mathrm{aff}}, \Delta Y^{\mathrm{aff}})$ by solving (46).
(2) Compute the predictor step length θ^{aff} and $(X^{\mathrm{aff}}, Y^{\mathrm{aff}})$.
(3) Compute the normalized duality gaps $\hat{\mu}$ and $\hat{\mu}^{\mathrm{aff}}$ by (50) and (52), respectively, centering parameter σ by (51) and barrier parameter τ by (49).
(4) Compute the predictor-corrector direction $(\Delta X, \Delta Y)$ by solving (53).
(5) Compute the step length θ and $(\bar{X}, \bar{Y})$.
(6) Set $(X,Y) \leftarrow (\bar{X}, \bar{Y})$.
(7) Let $r \leftarrow r+1$.
end while

Return X

Fig. 1. Predictor–corrector algorithm for the semidefinite least-squares problem (33).

6.1. Process description

The reactor is jacketed and excess heat is removed by an external heat exchanger. From mass and energy balances we get four differential equations:

$$\dot{c}_A = \frac{F}{V_\mathrm{R}}(c_{A0} - c_A) - k_1(T)c_A - k_3(T)c_A^2, \tag{57}$$

$$\dot{c}_B = -\frac{F}{V_\mathrm{R}}c_B + k_1(T)c_A - k_2(T)c_B, \tag{58}$$

$$\dot{T} = -\frac{1}{\rho C_p}\left(k_1(T)c_A\Delta H_{r1} + k_2(T)c_B\Delta H_{r2} + k_3(T)c_A^2\Delta H_{r3}\right) + \frac{F}{V_\mathrm{R}}(T_0 - T) + \frac{k_w A_R}{\rho C_p V_\mathrm{R}}(T_J - T), \tag{59}$$

$$\dot{T}_J = \frac{1}{m_J C_{pJ}}(\dot{Q}_J + k_w A_R(T - T_J)), \tag{60}$$

where c_A and c_B are the concentrations of components A and B in the effluent stream, respectively, T is the reaction temperature and T_J is the coolant temperature. c_{A0} is the concentration of A in the feed stream, F is the feed flow rate, V_R the reactor volume and $\dot{Q}_J$ is the rate of heat addition or removal.

The reaction coefficients k_1, k_2 and k_3 are given by Arrhenius's equation:

$$k_i(T) = k_{0i} \cdot \exp(E_i/T). \tag{61}$$

The model parameters are given in Table 2.

The model is a deterministic system of ordinary differential equations which is then linearized around the steady-state in Table 3, yielding

$$\dot{x}(t) = A_c x(t) + B_c u(t) + B_{pc} p(t) \tag{62}$$

Table 2
Model parameters

Symbol	Value	Symbol	Value
k_{01} (h^{-1})	1.287×10^{12}	ΔH_{r1} (kJ/mol)	4.2
k_{02} (h^{-1})	1.287×10^{12}	ΔH_{r2} (kJ/mol)	−11.0
k_{03} (l/(mol h))	9.043×10^{9}	ΔH_{r3} (kJ/mol)	−41.85
E_1 (K)	−9758.3	ρ (kJ/l)	9.9342
E_2 (K)	−9758.3	C_p (kJ/(kg K))	3.01
E_3 (K)	−8560	k_w (kJ/(h m^2 K))	4032
A_R (m^2)	0.215	V_R (m^3)	0.01
m_J (kg)	5.0	C_{pJ} (kJ/(kg K))	2.0

Table 3
Nominal steady-state values for Van der Vusse CSTR

Symbol	Value
c_A^s (mol/l)	2.1404
c_B^s (mol/l)	1.0903
T^s (K)	387.34
T_J^s (K)	386.06
F^s (l/h)	141.9
$\dot{Q}_J^s$ (kJ/h)	−1113.5
c_{A0}^s (mol/l)	5.1
T_0^s (K)	378.05

in which

$$x = \begin{bmatrix} c_A - c_A^s \\ c_B - c_B^s \\ T - T^s \\ T_J - T_J^s \end{bmatrix}, \quad u = \begin{bmatrix} F - F^s \\ \dot{Q}_J - \dot{Q}_J^s \end{bmatrix}, \quad p = \begin{bmatrix} c_{A0} - c_{A0}^s \\ T_0 - T_0^s \end{bmatrix}.$$

Model (62) is augmented with a stochastic term to describe the random part of the state evolution. This gives the following form of stochastic differential equations:

$$\mathrm{d}x(t) = (A_c x(t) + B_c u(t) + B_{pc} p(t))\mathrm{d}t + \sigma\,\mathrm{d}\omega(t) \tag{63}$$

in which $\{\omega(t)\}$ is a standard Wiener process. We also assume that the concentration of component B, c_B and the reaction temperature T are measured

$$y(t) = Cx(t) + v(t). \tag{64}$$

We select $\sigma = 0.03\,\mathrm{diag}([c_A^s, c_B^s, T^s, T_J^s])$ and

$$R_v = \begin{bmatrix} 0.0033 & 0 \\ 0 & 1.0 \end{bmatrix}.$$

The model is discretized with sample time $T_s = 0.01$ h. The process noise covariance for the discrete-time system is obtained as [12]

$$Q_w = \mathrm{e}^{A_c T_s}\left(\int_0^{T_s} \mathrm{e}^{-A_c\tau}\sigma\sigma^{\mathrm{T}}\mathrm{e}^{-A_c^{\mathrm{T}}\tau}\,\mathrm{d}\tau\right)\mathrm{e}^{A_c^{\mathrm{T}} T_s}.$$

The control objective is to control the concentration of B by manipulating the feed flow rate F and the rate of heat addition $\dot{Q}_J$. The concentration of B, c_B, and the temperature T in the reactor are available from measurements. The feed concentration c_{A0} and the feed temperature T_0 are treated as unmeasured disturbances.

6.2. LQG controller

We seek to solve the following optimization problem:

$$\min_{\{u_k\}} \quad \frac{1}{2}\sum_{k=1}^{\infty}\Big\{(z_k - \bar{z}_k)^{\mathrm{T}} Q_z (z_k - \bar{z}_k) + (u_k - u_t)^{\mathrm{T}} Q_u (u_k - u_t)\Big\} \tag{65}$$

$$\begin{aligned} \text{subject to} \quad & \bar{x}_{k+1} = \Phi_x \bar{x}_k + \Phi_u u_k + \Phi_d d_k + w_k, \\ & d_{k+1} = d_k + \xi_k, \\ & z_k = \Gamma_x \bar{x}_k + \Gamma_d d_k, \\ & y_k = z_k + v_k \end{aligned} \tag{66}$$

and where $\bar{z}_k$ is the setpoint for z_z and u_t is the target for u_k. This can be viewed as an unconstrained model predictive control problem with infinite horizon. A linear quadratic Gaussian controller is used instead of MPC to control the process. The optimal control law is given by

$$u_k = -L(\hat{x}_{k|k-1} - x_t) + u_t, \tag{67}$$

where the controller gain L is solved from

$$L = (\Phi_u^{\mathrm{T}} S \Phi_u + Q_u)^{-1}\Phi_u^{\mathrm{T}} S \Phi_x \tag{68}$$

and S is the solution to the discrete algebraic Riccati equation

$$S = \Phi_x^{\mathrm{T}} S \Phi_x + \Gamma_x^{\mathrm{T}} Q_z \Gamma_x - \Phi_x^{\mathrm{T}} S \Phi_u (\Phi_u^{\mathrm{T}} S \Phi_u + Q_u)^{-1}\Phi_u^{\mathrm{T}} S \Phi_x. \tag{69}$$

For setting the target values an augmented model is considered. The target values x_t and u_t are computed from [20,7]

$$\begin{bmatrix} I - \Phi_x & -\Phi_u \\ \Gamma_x & 0 \end{bmatrix}\begin{bmatrix} x_t \\ u_t \end{bmatrix} = \begin{bmatrix} \Phi_d \hat{d} \\ \bar{z} - \Gamma_d \hat{d} \end{bmatrix}. \tag{70}$$

The target vectors x_t and u_t can be expressed explicitly as functions of $\bar{z}$ and $\hat{d}$ by solving (70) [7]

$$\begin{bmatrix} x_t \\ u_t \end{bmatrix} = T_y \bar{z} + T_d \hat{d}. \tag{71}$$

6.3. Kalman filter tuning

The linear stochastic discrete-time system was used as the true system in the closed-loop simulations. Process noise and measurement noise were added. A Kalman filter based on initial noise covariances $Q_{w,0} = I$, $Q_{\xi,0} = I$, $R_{v,0} = \mathrm{diag}(0.1, 0.3)$ and $S_{wv,0} = 0$ was designed. The disturbance model used in the LQG controller was a pure input disturbance, i.e. $\Phi_d = \Phi_x$, $\Gamma_d = 0$.

The weight matrices Q_z and Q_u for the LQG controller were selected as

$$Q_z = \begin{bmatrix} 100 & 0 \\ 0 & 1 \end{bmatrix}, \quad Q_u = \begin{bmatrix} 0.1 & 0 \\ 0 & 10^{-5} \end{bmatrix}.$$

The weights were chosen such that the contribution from the signals would be of the same magnitude but with less weight on inputs than outputs.

The autocovariance least-squares method was used to estimate the noise covariances, which were then used to compute a new Kalman filter gain. An unmeasured disturbance in the form of the feed concentration c_{A0} was introduced. The disturbance was modelled as an integrator driven by white noise with covariance 0.001. A sequence of $N_{\mathrm{d}} = 1000$ measurements, corresponding to 10 h, was used to tune the filter. The tuned filter was then tested on another 1000 data points. The regularization parameters were set to $\lambda = 1$, $\rho = 0$ and a lag of $L = 15$ was used when computing the autocovariances.

A performance measure was defined for comparing different filter tunings:

$$\overline{\Phi} = \frac{1}{N_{\mathrm{d}}}\sum_{k=1}^{N_{\mathrm{d}}}\Big\{(z_k - \bar{z}_k)^{\mathrm{T}} Q_z (z_k - \bar{z}_k)(u_k - u_u)^{\mathrm{T}} Q_u (u_k - u_t)\Big\}. \tag{72}$$

The improvement is measured as the improvement in the performance measure

$$\eta = \frac{\overline{\Phi}_{\text{original tuning}} - \overline{\Phi}_{\text{new tuning}}}{\overline{\Phi}_{\text{original tuning}}}. \tag{73}$$

Furthermore, the root-mean square error of output predictions

$$V_{\mathrm{RMS}} = \sqrt{\frac{1}{N_{\mathrm{d}}}\sum_{k=1}^{N_{\mathrm{d}}}\|y_k - \Gamma_x \hat{x}_{k|k-1} - \Gamma_d \hat{d}_{k|k-1}\|_2^2} \tag{74}$$

was computed. The Kalman filter tuning experiments were repeated 200 times using different realizations of the noise. The improvement in the cost function is shown in Fig. 2 and the root-mean square error of the output predictions is depicted in Fig. 3. The ideal Kalman filter, where perfect

knowledge of both stochastic and deterministic parts of the plant is assumed, is included for comparison. By using the tuned filter, the performance measure $\overline{\Phi}$ was improved by an average of 17.7%, compared to 25.0% with the ideal filter. The root-mean square error of the output predictions (74) was improved on average by 9.2% using tuning and by 11.2% using the ideal filter.

An additional set of simulations illustrate the change in performance due to filter tuning. In the simulations the process started at steady-state and at $k = 500$ samples a

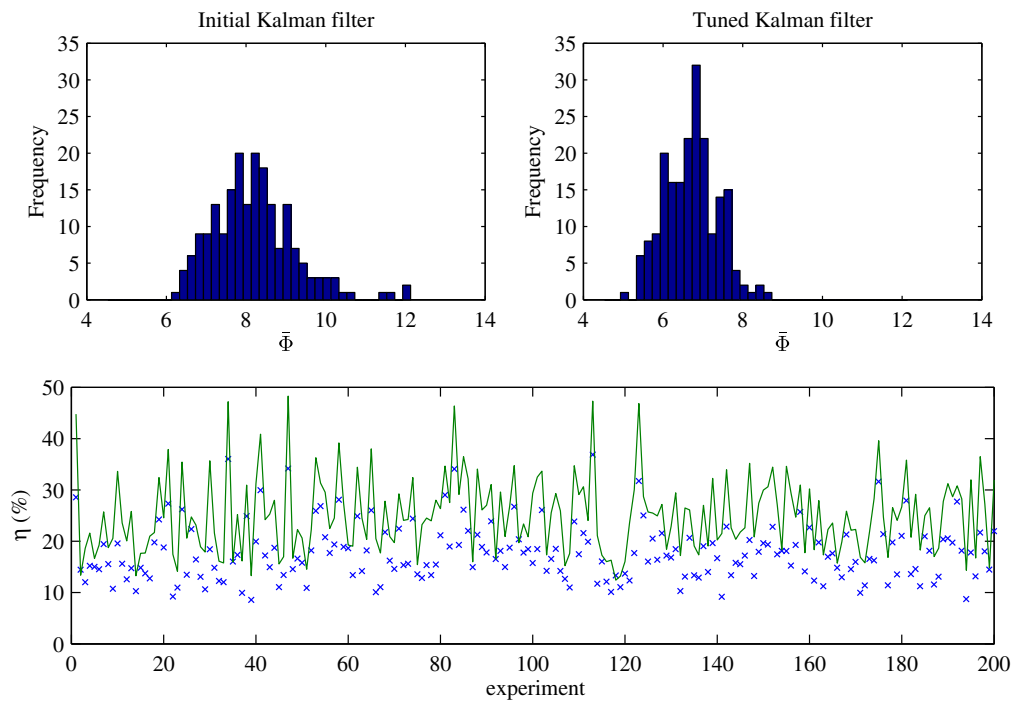

Fig. 2. *Upper graphs*: Histograms of performance measure (72) based on 200 experiments. *Lower graph*: Improvement in the performance measure using the tuned Kalman filter (crosses) and the ideal Kalman filter (solid line).

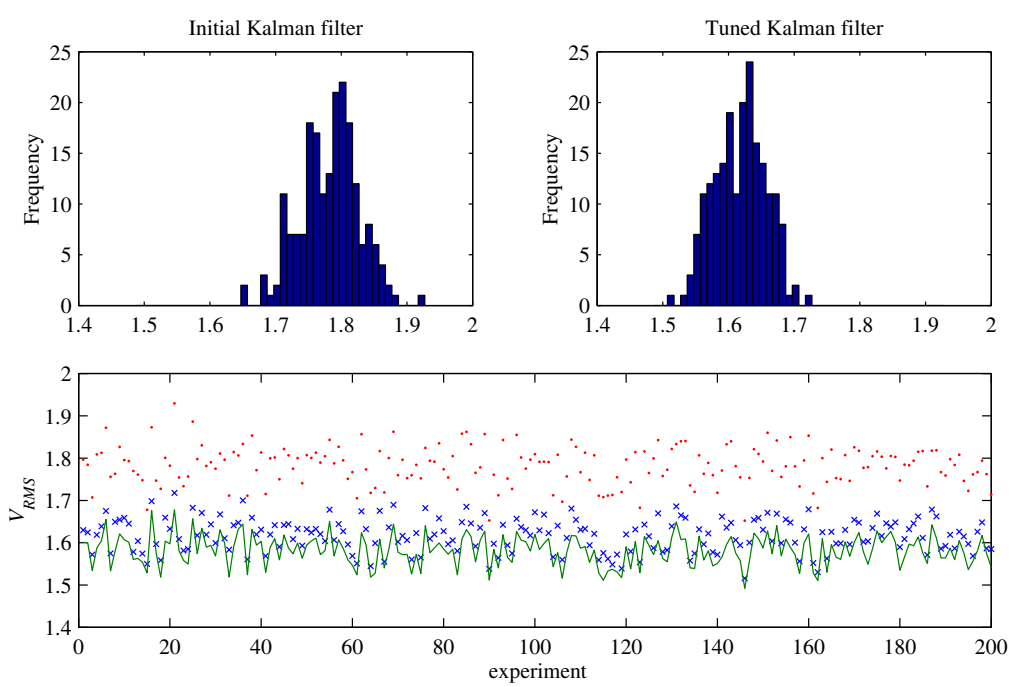

Fig. 3. *Upper graphs*: Histograms of root-mean square error of output predictions (74) based on 200 experiments. *Lower graph*: Root-mean square error of output predictions using the initial Kalman filter (dots), the tuned Kalman filter (crosses) and the ideal Kalman filter (solid line).

step change was made in setpoint $\bar{z}_1$ to 0.29 (corresponding to $c_B = 1.38$ mol/l). A drifting disturbance like the one used in the tuning experiments was also included. Responses are shown in Figs. 4 and 5. The simulations were performed using identical noise sequences. An improvement in setpoint tracking after tuning is observed from Fig. 4. It is

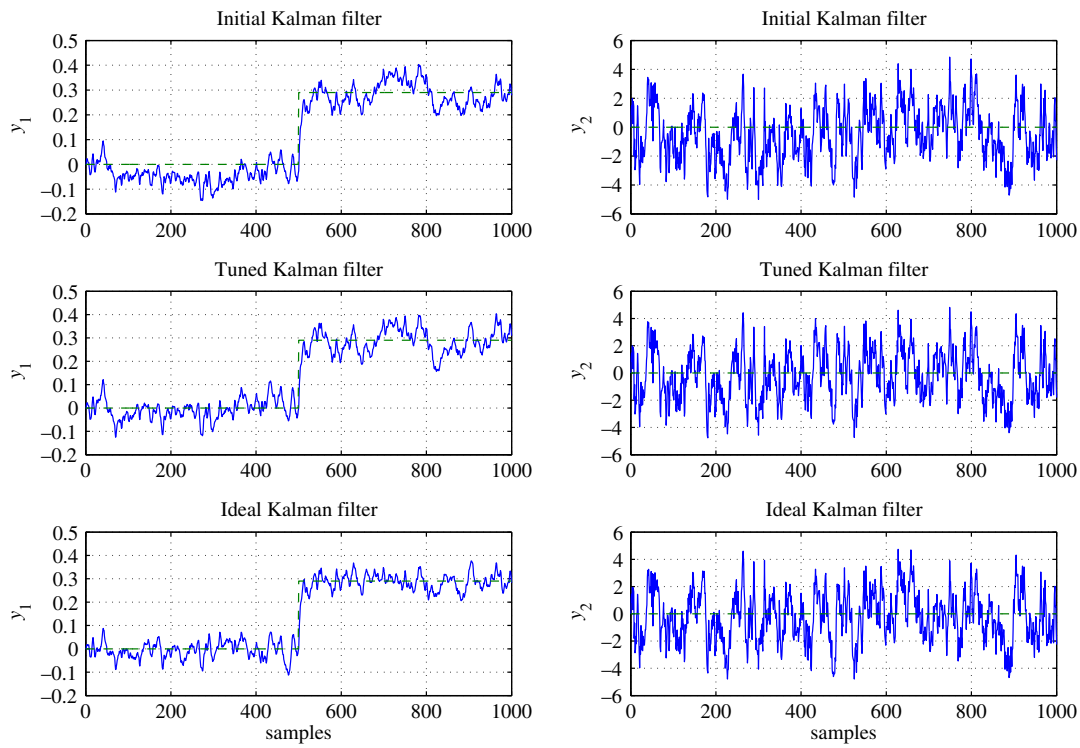

Fig. 4. Comparison of output responses to a setpoint change at 500 samples. The measurement noise has been removed for clarity.

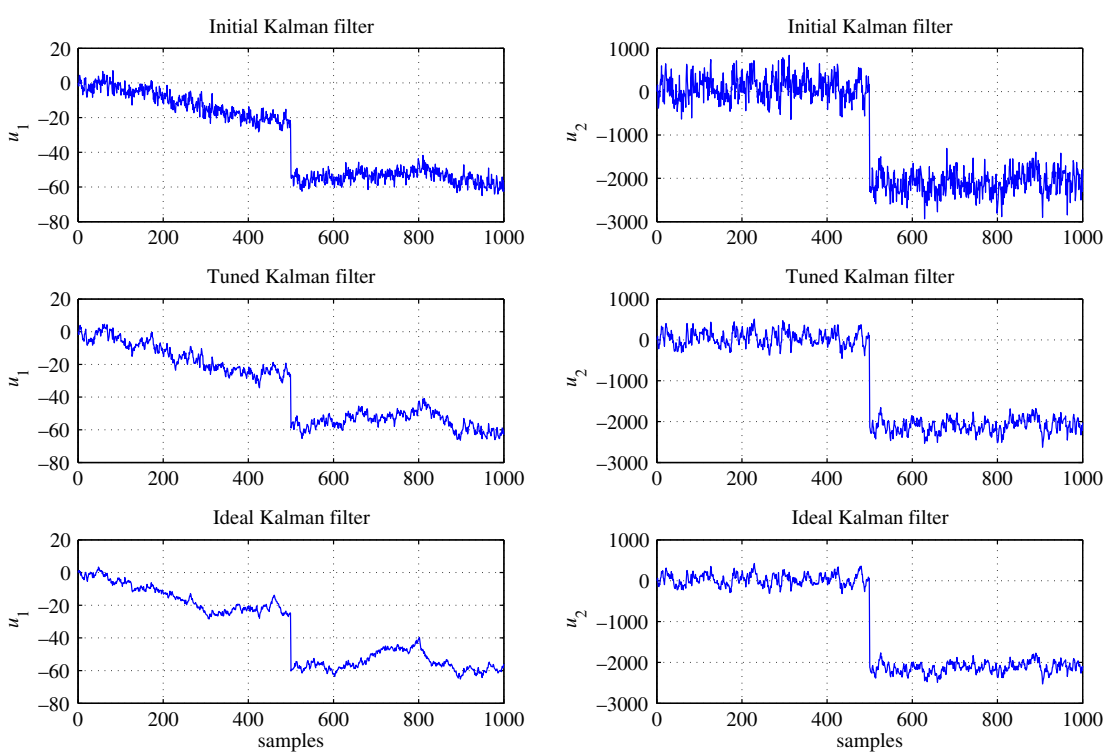

Fig. 5. Comparison of input responses to a setpoint change at 500 samples.

also interesting to note from Fig. 5 that the variance in the control inputs decreases as the Kalman filter tuning is improved.

7. Conclusions

A method for Kalman filter tuning was presented. The method used is a generalization of the autocovariance least-squares method to systems with mutually correlated noise. Regularization was included in order to handle ill-conditioning of the least-squares problem. The results show that significant improvement in the control performance can be achieved with the tuning method.

The symmetric positive semidefinite least-squares problem arising from the covariance estimation problem can be solved by an interior-point method. A predictor–corrector method was presented.

Acknowledgements

The authors would like to thank Murali Rajamani and Professors James B. Rawlings and Per Christian Hansen for fruitful discussions.

Appendix A. Explicit solution of the Lyapunov equation

From (13) we get the following Lyapunov equation:

$$P_{\mathrm{p}} = \overline{A} P_{\mathrm{p}} \overline{A}^{\mathrm{T}} + \overline{G} E[\bar{w}_k \bar{w}_k^{\mathrm{T}}] \overline{G}^{\mathrm{T}}. \tag{A.1}$$

Since

$$E[\bar{w}_k \bar{w}_k^{\mathrm{T}}] = \begin{bmatrix} Q_w & 0 & S_{wv} \\ 0 & Q_\xi & 0 \\ S_{wv}^{\mathrm{T}} & 0 & R_v \end{bmatrix}$$

and

$$\overline{G} = \begin{bmatrix} \begin{bmatrix} G & 0 \\ 0 & I \end{bmatrix} & -K_{\mathrm{p}} \end{bmatrix},$$

we can expand the Lyapunov equation to

$$\begin{aligned} P_{\mathrm{p}} = {} & \overline{A} P_{\mathrm{p}} \overline{A}^{\mathrm{T}} + V_x Q_w V_x^{\mathrm{T}} + V_d Q_\xi V_d^{\mathrm{T}} - K_{\mathrm{p}} S_{wv}^{\mathrm{T}} V_x^{\mathrm{T}} \\ & - V_x S_{wv} K_{\mathrm{p}}^{\mathrm{T}} + K_{\mathrm{p}} R_v K_{\mathrm{p}}^{\mathrm{T}}. \end{aligned} \tag{A.2}$$

By applying the vec operator to (A.2), the equation can be solved for P_{p}

$$\begin{aligned} \mathrm{vec}(P_{\mathrm{p}}) = {} & (I_{(n_x+n_d)^2} - \overline{A} \otimes \overline{A})^{-1} ((V_x \otimes V_x)\mathrm{vec}(Q_w) \\ & + (V_d \otimes V_d)\mathrm{vec}(Q_\xi) - (V_x \otimes K_{\mathrm{p}})\mathrm{vec}(S_{wv}^{\mathrm{T}}) \\ & - (K_{\mathrm{p}} \otimes V_x)\mathrm{vec}(S_{wv}) + (K_{\mathrm{p}} \otimes K_{\mathrm{p}})\mathrm{vec}(R_v)). \end{aligned}$$

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